

Qassim university

College of Computer

Department of computer science



CS471 – Web Technologies

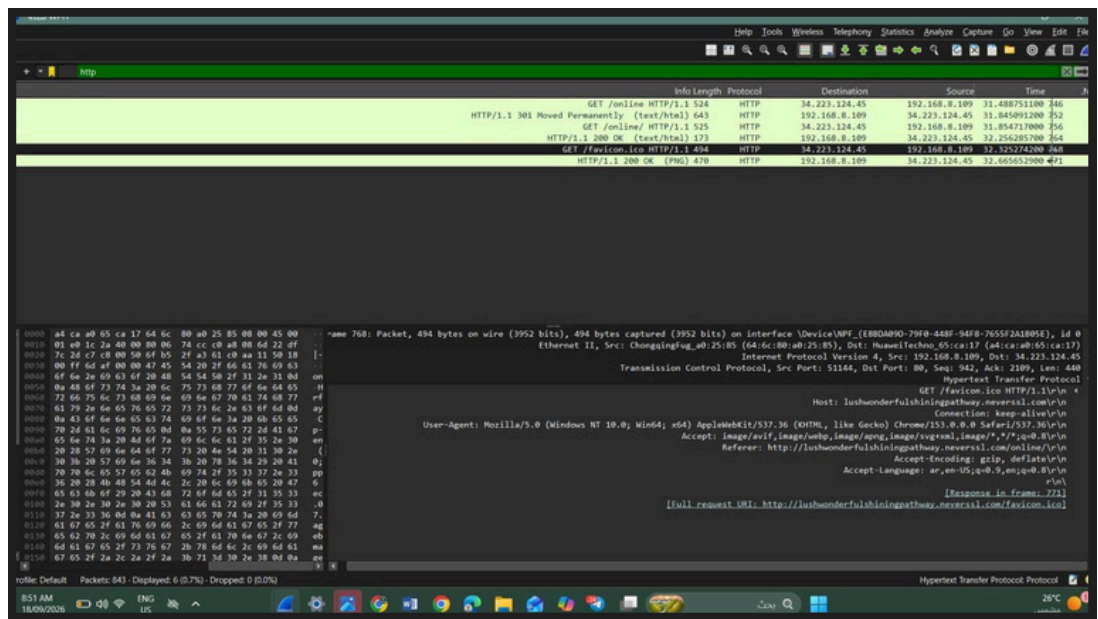
Assignment 1

**CS471 – Web Technologies
(Laboratory)**

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Part 1 Capturing HTTP Traffic

Task 1 & Task 2: Filter and Analyze HTTP Packets

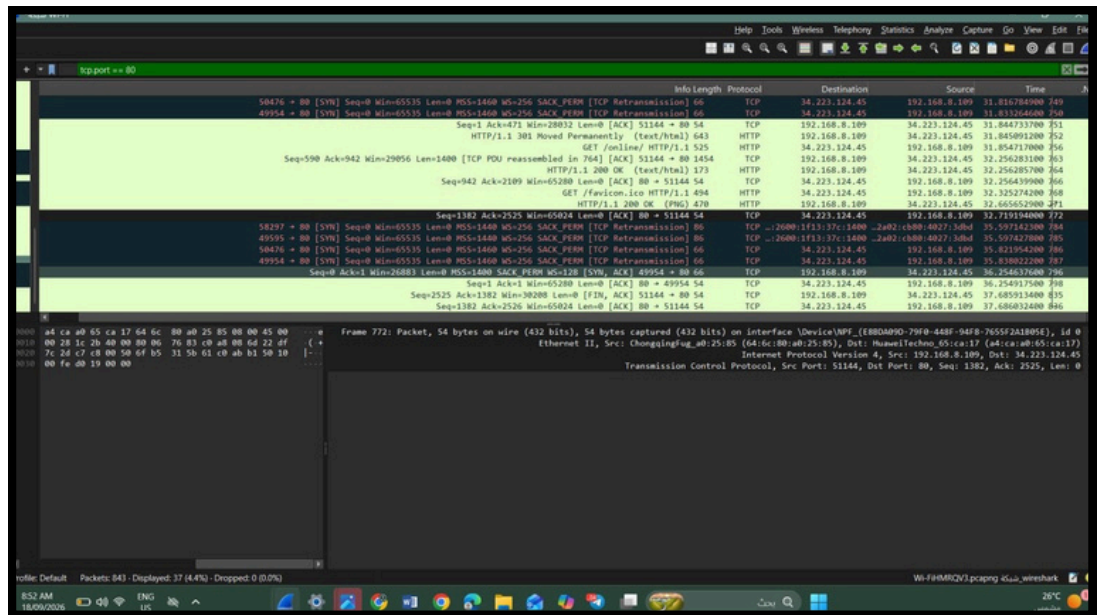


Captured unencrypted HTTP traffic from neverssl.com.

And Applied the http filter to inspect requests and responses. Observed HTTP request methods like GET /online and response status codes such as 200 OK and 301 Moved Permanently. Standard HTTP traffic transfers plain-text data without encryption.

Part 2: Analyzing TCP/IP Traffic

Task 1: Filter TCP Packets



Filtered packets using tcp.port == 80 and used the Follow -> TCP Stream feature to view the entire stream conversation between the client (192.168.8.109) and the server (34.223.124.45).

Task 2: Analyze TCP Handshake, Data Transfer, and Termination

50476 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM [TCP Retransmission] 66	TCP	34.223.124.45	192.168.8.109	35.821954200	786
49954 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM [TCP Retransmission] 66	TCP	34.223.124.45	192.168.8.109	35.838022200	787
Seq=0 Ack=1 Win=26883 Len=0 MSS=1400 SACK_PERM WS=128 [SYN, ACK] 49954 → 80 66	TCP	192.168.8.109	34.223.124.45	36.254637600	796
Seq=1 Ack=1 Win=65280 Len=0 [ACK] 80 → 49954 54	TCP	34.223.124.45	192.168.8.109	36.254917500	798

Three-Way Handshake: Connection established using SYN (client request), SYN-ACK (server response), and ACK (client acknowledgment) packets.

HTTP/1.1 301 Moved Permanently (text/html) 643	HTTP	192.168.8.109	34.223.124.45	31.845091200	752
GET /online/ HTTP/1.1 525	HTTP	34.223.124.45	192.168.8.109	31.854717000	756
Seq=590 Ack=942 Win=29056 Len=1400 [TCP PDU reassembled in 764] [ACK] 51144 → 80 1454	TCP	192.168.8.109	34.223.124.45	32.256283100	763
HTTP/1.1 200 OK (text/html) 173	HTTP	192.168.8.109	34.223.124.45	32.256285700	764
Seq=942 Ack=2109 Win=65280 Len=0 [ACK] 80 → 51144 54	TCP	34.223.124.45	192.168.8.109	32.256439900	766

Data Transfer: Observed HTTP requests/responses exchanged using Sequence (Seq) and Acknowledgment (Ack) numbers to maintain packet order and delivery integrity.

3 Len=0 MSS=1400 SACK_PERM WS=128 [SYN, ACK] 49954 → 80 66	TCP	192.168.8.109	34.223.124.45	36.254637600	796
Seq=1 Ack=1 Win=65280 Len=0 [ACK] 80 → 49954 54	TCP	34.223.124.45	192.168.8.109	36.254917500	798
Seq=2525 Ack=1382 Win=30208 Len=0 [FIN, ACK] 51144 → 80 54	TCP	192.168.8.109	34.223.124.45	37.685913400	835
Seq=1382 Ack=2526 Win=65024 Len=0 [ACK] 80 → 51144 54	TCP	34.223.124.45	192.168.8.109	37.686032400	836

TCP Termination: Identified connection closure handled gracefully via FIN, ACK and final ACK packets.

Part 3: Capturing and Analyzing UDP Traffic

Task 1 & Task 2: Filter and Analyze UDP Packets

Len=29 443 → 60488 71	UDP	216.239.38.120	192.168.8.109	3.425488300	19
Len=29 443 → 60488 71	UDP	216.239.38.120	192.168.8.109	3.441928100	20
Len=23 60488 → 443 65	UDP	192.168.8.109	216.239.38.120	3.614753200	22
Len=24 60488 → 443 66	UDP	192.168.8.109	216.239.38.120	3.614929200	23
Len=24 60488 → 443 66	UDP	192.168.8.109	216.239.38.120	3.614930700	24
Len=1246 443 → 60488 1288	UDP	216.239.38.120	192.168.8.109	3.709679500	25
Len=825 443 → 60488 867	UDP	216.239.38.120	192.168.8.109	3.709921500	26
Len=27 60488 → 443 69	UDP	192.168.8.109	216.239.38.120	3.828665800	28
Len=201 443 → 60488 243	UDP	216.239.38.120	192.168.8.109	3.848920500	29
G, CRYPTO, CRYPTO, CRYPT 1292	QUIC	216.239.38.120	192.168.8.109	3.916086500	31
66708c802fa59e6, PKN: 2, CRYPTO 1292	QUIC	216.239.38.120	192.168.8.109	3.916351900	32
RYPTO, CRYPTO, PING, CRYPTO, CR 1292	QUIC	216.239.38.120	192.168.8.109	3.916493500	33

68 bits), 71 bytes captured (568 bits) on interface \Device\NPF_{E8BDA09D-79F0-448F-94F8-7655F2A1B05E}, id 0

Ethernet II, Src: ChongqingFug_a0:25:85 (64:6c:80:a0:25:85), Dst: HuaweiTechno_65:ca:17 (a4:ca:a0:65:ca:17)

Internet Protocol Version 4, Src: 192.168.8.109, Dst: 216.239.38.120

User Datagram Protocol, Src Port: 60488, Dst Port: 443

Source Port: 60488

Destination Port: 443

Length: 37

Checksum: 0x9b50 [unverified]

[Checksum Status: Unverified]

[Stream index: 0]

[Stream Packet Number: 9]

[Timestamps]

UDP payload (29 bytes)

Data (29 bytes)

Applied the udp filter to inspect connectionless datagrams. Analyzed UDP headers which consist of only Source Port (60488), Destination Port (443), Length (37), and Checksum. Compared to TCP, UDP headers are simple, lightweight, and lack sequence numbers or connection establishment steps.

Part 4: Comparing TCP and UDP

Task 1: Reliability and Connection Establishment

Criterion	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	TCP is connection-oriented, establishing connections via a 3-way handshake (SYN, SYN-ACK, ACK). It guarantees packet delivery through acknowledgments and retransmissions. UDP is connectionless and doe
Data Integrity and Ordering	TCP	TCP uses sequence numbers to ensure packets are reassembled in the correct order and performs error checking to retransmit lost or corrupted packets. UDP sends datagrams independently without sequence tracking or ordering

Task 2: Identify the Use Cases and Performance of TCP and UDP

Feature	TCP	UDP
Uses Cases	Web Browsing (HTTP/HTTPS), File Transfers (FTP), Email (SMTP/IMAP), Remote Access (SSH).	Video Streaming, Online Multiplayer Gaming, VoIP calls, DNS queries.
Performance	Slower Overhead: Higher latency due to connection handshake, 20-byte minimum header size, flow control, and congestion control algorithms.	Faster & Low Latency: Fast transmission with no connection setup and a minimal 8-byte header size.